Claim

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$\langle\cdot|\cdot\rangle$ is an inner product on V and $x, y \in V$

$$\Downarrow$$

$$|\langle x|y\rangle|^2 - \|x\|^2\|y\|^2 = \frac{1}{4}\left(|\langle x+y|x-y\rangle|^2 - \|x+y\|^2\|x-y\|^2\right)$$

$$\Downarrow$$

$$|\langle x|y\rangle|^2 \leq \frac{1}{4}|\langle x+y|x-y\rangle|^2 + \|x\|^2\|y\|^2$$

$$\Downarrow$$

$$|\langle x|y\rangle|^2 \leq \|x\|^2\|y\|^2$$